

Mathematical Solutions to Fundamental Physics Problems with T0 Theory Part 1

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Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Gravitational constant: $G = 1/(\xi E_p^2) \rightarrow G = 1/E_p^2$ (factor 7500), derivation from ξ via Doc. 012.
- Further corrections: Dirac condition $q_m = \frac{1}{4\pi} \oint \vec{B} \cdot d\vec{A} \rightarrow \oint \vec{B} \cdot d\vec{A}$ (consistent with $2\pi n\hbar$); $\pi_2(S^1) = \mathbb{Z} \rightarrow \pi_1(U(1)) = \mathbb{Z}$; $m_e = \xi E_0^2$ with $E_0 = 0.511 \text{ MeV} \rightarrow$ lepton ladder (Doc. 006); neutrino: Δm_{21}^2 only in order of magnitude (factor 3.6). Noted: extra factor $1/L_0$ in θ_L unnecessary (makes θ_L dimensionful); equivalence proof circular; conformally flat metric gives light deflection 0; prefactor $1/\Lambda^2$ in the Riemann tensor; $E_{\text{char}} = \hbar$ gives $q_e = \xi$; $\nabla E^R = -\nabla E^L$ does not make $\mathcal{L}_{\text{weak}}$ vanish.
- Prediction table (1 Oct. 2026): Δm_{21}^2 ✓Consistent $\rightarrow$ order of magnitude (factor 3.6); Λ ✓Exact $\rightarrow$ not reproduced; H_0 $0.7\sigma \rightarrow 2.4\sigma/6.8\sigma$.
- Noted (1 Oct. 2026): cosmological constant $\frac{\lambda}{4} \langle E_{\text{field}} \rangle^4$ gives, with $\lambda = 2\xi^2$ and $\langle E_{\text{field}} \rangle = \nu$, about $(2.4 \text{ GeV})^4$, not $(10^{-3} \text{ eV})^4$.

Abstract

T0 Theory: An Elegant Mathematical Solution to the Three Major “Uglinesses” of the Standard Model and Gravity

The T0 theory, with its single fundamental parameter $\xi = \frac{4}{3} \times 10^{-4}$ and the universal energy field $E_{\text{field}}(x, t)$, solves three central aesthetic and structural problems of modern physics in the most natural way:

1. **Chirality** becomes a geometric consequence of the rotation direction of the energy field: $\text{Chirality} = \text{sgn}(\nabla \times \vec{E}_{\text{field}})$. The exclusive left-handedness of the weak interaction emerges without additional assumptions.

2. **Gravity** is not a separate tensor term but the gradient of the same energy field. The nonlinear field equation $\square E_{\text{field}} + \xi E_{\text{field}}^3 = 0$ is mathematically equivalent to Einstein's theory of gravity (proven in the weak-field limit and through complete covariant tensor formulation $g_{\mu\nu}(E_{\text{field}})$ including Riemann and Ricci tensors).

3. **Magnetic Monopoles** exist as topological excitations of the energy field and satisfy exactly the Dirac quantization condition $q_e q_m = 2\pi n \hbar$. Their rarity is a natural consequence of the high energy threshold $\sim E_P/\xi$.

The theory is fully covariant, renormalizable, canonically quantizable, and contains the Standard Model as an effective low-energy theory. All couplings, masses, and cosmological parameters (including the fine structure constant α , the muon g-2 anomaly, the cosmological constant Λ_{cosmo} , and the Hubble tension) are derived from the single fundamental parameter ξ and the fractal geometry of T0 cells; the derivation is still under development and individual steps remain speculative.

Thus it is shown: Physics is not "ugly" – it only becomes beautiful when derived from a single principle.

1. Chirality – The Left-Right Asymmetry

The Problem

Particles exist in left- and right-handed versions with different behavior – an "ugly" asymmetry without explanation.

T0 Solution: Energy Field Rotation

Fundamental insight: Chirality arises from the **rotation direction of the energy field** $E_{\text{field}}(x, t)$.

Mathematical Derivation

Left-handed particles:

$$E_{\text{field}}^L(x, t) = E_0 \cdot e^{i(\omega t - \vec{k} \cdot \vec{x} + \theta_L)}$$

where the phase is:

$$\theta_L = +\frac{\xi}{2} \int (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A}$$

Right-handed particles:

$$E_{\text{field}}^R(x, t) = E_0 \cdot e^{i(\omega t - \vec{k} \cdot \vec{x} + \theta_R)}$$

where:

$$\theta_R = -\frac{\xi}{2} \int (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A}$$

The Geometric Explanation

Chirality = Sign of the Energy Field Rotation:

$$\text{Chirality} = \text{sgn}(\nabla \times \vec{E}_{\text{field}})$$

Left-handed: $\nabla \times \vec{E}_{\text{field}} > 0$ (right-handed rotation)

Right-handed: $\nabla \times \vec{E}_{\text{field}} < 0$ (left-handed rotation)

Why Weak Interaction Couples Only to Left-Handed Particles

The weak interaction couples to the **gradient of the energy field**:

$$\mathcal{L}_{\text{weak}} = \xi^{1/2} \cdot E_{\text{field}}^L \cdot \nabla E_{\text{field}}^L$$

This is non-zero only for **one chirality** because:

$$\nabla E_{\text{field}}^R = -\nabla E_{\text{field}}^L$$

(Note of 30 Sept. 2026: From $\nabla E^R = -\nabla E^L$ it follows that $E^R \nabla E^R = -E^R \nabla E^L$, not 0; a sign change does not make the term vanish. Moreover, E^L and E^R above differ only in the phase $\theta_{L,R}$, which does not yield $\nabla E^R = -\nabla E^L$.)

Result: The "ugly" chirality becomes the **natural consequence of 3D space geometry**.

2. Gravity & Standard Model – The Ungraceful Integration

The Problem

The curvature of spacetime ($R_{\mu\nu}R^{\mu\nu}$) does not fit elegantly with the other forces.

T0 Solution: Gravity as Energy Field Gradient

Fundamental insight: Gravity is **not a separate force** but the **gradient of the universal energy field**.

Einstein's Field Equations Reinterpreted

Standard GR:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G T_{\mu\nu}$$

T0 Energy Field Form:

$$\nabla^2 E_{\text{field}} = 4\pi G \rho_E \cdot E_{\text{field}}$$

This **Poisson-like equation** for energy replaces the complex tensor structure!

Connection to the Metric

The spacetime metric arises from the energy field:

$$g_{\mu\nu} = \eta_{\mu\nu} \cdot \left(1 - \frac{2\xi \cdot E_{\text{field}}}{E_P} \right)$$

where $\eta_{\mu\nu}$ is the Minkowski metric. (Note of 30 Sept. 2026: This metric is purely conformally flat. Null geodesics are conformally invariant, so light propagates as in flat space: light deflection 0 (Nordström case) instead of the measured 1.75'' at the solar limb ($\gamma = 1$ to $\sim 10^{-4}$). As a complete metric it is therefore not equivalent to GR; this also applies to the equivalence statements in the abstract and in Section 2 of the reply.)

Unified Lagrangian

All forces + Gravity:

$$\mathcal{L}_{\text{total}} = \xi \cdot (\partial E_{\text{field}})^2$$

That's it! A single Lagrangian for:

- Electromagnetism
- Weak interaction
- Strong interaction
- **Gravity**

The "squared curvature" disappears – replaced by **squared energy field gradients**.

Gravitational Constant Derived

$$G = \frac{1}{E_p^2}, \quad G_{\text{SI}} = \frac{\xi^2}{4m_e} C_{\text{conv}} K_{\text{frak}} \text{ (Doc. 012)}$$

(Corrected on 30 Sept. 2026: previously $G = 1/(\xi E_p^2)$ – in natural units $G = 1/E_p^2$; the additional factor $1/\xi$ made G too large by 7500. The derivation from ξ goes via Doc. 012.)

Result: Gravity becomes just as "pretty" as the other forces.

3. Magnetic Monopoles – The Hidden Symmetry

The Problem

Maxwell's equations would be more symmetric with magnetic monopoles, but they don't seem to exist.

T0 Solution: Emergent Symmetry from Energy Field Topology

Standard Maxwell Equations (asymmetric)

$$\nabla \cdot \vec{E} = \rho / \epsilon_0 \quad (\text{electric charge exists})$$

$$\nabla \cdot \vec{B} = 0 \quad (\text{no magnetic charge})$$

T0 Energy Field Interpretation

Electric charge = Localized energy field source:

$$q_e = \int E_{\text{field}} d^3x$$

Magnetic field = Rotation of the energy field:

$$\vec{B} = \nabla \times \vec{A} = \nabla \times (E_{\text{field}} \cdot \hat{n})$$

Why No Magnetic Monopoles?

Topological condition:

$$\oint \vec{B} \cdot d\vec{A} = \oint (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A} = 0$$

This holds **always** by Stokes' theorem because the energy field E_{field} is **globally defined**.

The Hidden Symmetry Revealed

The **true symmetry** is not electric-magnetic, but:

$$\text{Energy field source} \leftrightarrow \text{Energy field rotation}$$

Mathematically:

$$\text{Electric: } \nabla \cdot E_{\text{field}} = \rho_E$$

$$\text{Magnetic: } \nabla \times E_{\text{field}} = \vec{j}_E$$

This **is perfectly symmetric** in energy field space!

Why We Don't See Monopoles

In the 3D projection, this symmetry appears broken because:

$$\vec{B}_{\text{observed}} = \text{Projection}(\nabla \times E_{\text{field}})$$

The symmetry is **not hidden** – it exists at the fundamental energy field level but appears asymmetric in our macroscopic electromagnetic description.

Result: The “missing symmetry” is in fact **fully present** at the T0 energy field level.

The Ultimate Unification

All three “ugly” aspects vanish when we recognize:

$$\text{All physics} = \text{Geometry of the universal energy field } E_{\text{field}}(x, t)$$

With **one equation:**

$$\square E_{\text{field}} = 0$$

And **one parameter:**

$$\xi = \frac{4}{3} \times 10^{-4}$$

Physics becomes beautiful.

1. Chirality – Dimensional Analysis Corrected

Objection

“ $\theta_L = +\frac{\xi}{2} \int (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A}$ is dimensionally inconsistent”

CORRECT TO FORMULATION

The correct, dimensionally consistent formulation is:

$$\theta_L = +\frac{\xi}{2E_p} \int (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A}$$

where:

- ξ : dimensionless coupling parameter

- E_p : Planck energy (dimension Energy)
- $\vec{E}_{\text{field}}$: field strength (dimension Energy/Length)
- $d\vec{A}$: area element (dimension Length²)

Dimensional analysis:

$$\begin{aligned} [\theta_L] &= \frac{1}{E} \cdot \left[\frac{E}{L} \right] \cdot L^2 \\ &= \frac{E}{E} \cdot L = 1 \cdot L \end{aligned}$$

Correction with additional factor $1/L_0$ (characteristic length):

$$\theta_L = + \frac{\xi}{2E_p L_0} \int (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A}$$

Now: $[\theta_L] = \frac{1}{EL} \cdot \frac{E}{L} \cdot L^2 = 1 \checkmark$ dimensionless. (Note of 30 Sept. 2026: The dimensional analysis omits the $1/L$ of the curl: $[\nabla \times \vec{E}_{\text{field}}] = (E/L)/L = E/L^2$, times L^2 gives E , divided by E_p dimensionless. Already $\theta_L = \xi/(2E_p) \int \dots$ is dimensionless; the extra factor $1/L_0$ is what gives θ_L the dimension $1/L$.)

2. Gravity – Equivalence to Einstein Demonstrated

Objection

" $\nabla^2 E_{\text{field}} = 4\pi G \rho_E E_{\text{field}}$ is not equivalent to Einstein's equations"

PROOF OF EQUIVALENCE

The T0 equation **IS** equivalent to Einstein in the weak-field limit:

Einstein's equations (weak field):

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad \text{with } |h_{\mu\nu}| \ll 1$$

Linearized:

$$\square h_{\mu\nu} - \partial_\mu \partial_\alpha h_\nu^\alpha - \partial_\nu \partial_\alpha h_\mu^\alpha + \partial_\mu \partial_\nu h = -16\pi G T_{\mu\nu}$$

In harmonic gauge (Lorentz gauge):

$$\square h_{\mu\nu} = -16\pi G \left(T_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} T \right)$$

T0 form with energy-momentum tensor:

I show that the T0 equation is equivalent via:

$$E_{\text{field}} \leftrightarrow h_{00} \quad (\text{time-time component of the metric})$$

Rigorous proof:

Step 1: T0 field equation in tensor form

$$\nabla^2 E_{\text{field}} = 4\pi G \rho_E \cdot E_{\text{field}}$$

Step 2: Identification with metric perturbation

$$h_{00} = - \frac{2\xi \cdot E_{\text{field}}}{E_p}$$

Step 3: Substituting into Einstein equation (00 component)

$$\nabla^2 h_{00} = -8\pi G T_{00} = -8\pi G \rho c^2$$

In natural units ($c = 1$):

$$\nabla^2 h_{00} = -8\pi G \rho_E$$

Step 4: Inserting T0 relation

$$\nabla^2 \left(-\frac{2\xi E_{\text{field}}}{E_P} \right) = -8\pi G \rho_E$$

$$\frac{2\xi}{E_P} \nabla^2 E_{\text{field}} = 8\pi G \rho_E$$

$$\nabla^2 E_{\text{field}} = \frac{4\pi G E_P}{\xi} \rho_E$$

Step 5: With $\rho_E = E_{\text{field}} \cdot \rho_0$ (energy density coupling):

$$\nabla^2 E_{\text{field}} = \frac{4\pi G E_P}{\xi} \rho_0 \cdot E_{\text{field}}$$

Normalization: $\rho_0 = \xi / E_P$ yields:

$$\boxed{\nabla^2 E_{\text{field}} = 4\pi G \rho_E \cdot E_{\text{field}}} \quad \checkmark$$

PROOF COMPLETE: T0 is equivalent to Einstein in the relevant limit. (Note of 30 Sept. 2026: Substituting in Step 5 gives $\frac{4\pi G E_P}{\xi} \cdot \frac{\xi}{E_P} \cdot E_{\text{field}} = 4\pi G E_{\text{field}}$ – the ρ_E in the box has then disappeared. At the same time ρ_E is used both as the source and as $\rho_0 E_{\text{field}}$; the argument is circular and does not establish the equivalence.)

3. Nonlinearity and Full Covariance

T0 Contains Nonlinearity

The complete T0 field equation is:

$$\boxed{\square E_{\text{field}} + \xi \cdot E_{\text{field}}^3 = 0}$$

The cubic term E_{field}^3 provides the **nonlinearity**!

Derivation from the Lagrangian:

$$\mathcal{L} = \xi \cdot (\partial_\mu E_{\text{field}})(\partial^\mu E_{\text{field}}) - \frac{\lambda}{4} E_{\text{field}}^4$$

Euler-Lagrange equation:

$$\frac{\partial \mathcal{L}}{\partial E_{\text{field}}} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu E_{\text{field}})} = 0$$

Calculating the terms:

$$\frac{\partial \mathcal{L}}{\partial E_{\text{field}}} = -\lambda E_{\text{field}}^3$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu E_{\text{field}})} = 2\xi \partial^\mu E_{\text{field}}$$

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu E_{\text{field}})} = 2\xi \partial_\mu \partial^\mu E_{\text{field}} = 2\xi \square E_{\text{field}}$$

Inserting into Euler-Lagrange:

$$-\lambda E_{\text{field}}^3 - 2\xi \square E_{\text{field}} = 0$$

$$\square E_{\text{field}} = -\frac{\lambda}{2\xi} E_{\text{field}}^3$$

With $\lambda/(2\xi) = \xi$:

$$\square E_{\text{field}} + \xi \cdot E_{\text{field}}^3 = 0$$

This is a **nonlinear Klein-Gordon equation** – mathematically equivalent to nonlinear GR!
Solution in weak field:

$$E_{\text{field}} = E_0 + \epsilon(x) \quad \text{with } |\epsilon| \ll |E_0|$$

$$\square \epsilon + 3\xi E_0^2 \epsilon = 0 \quad (\text{linearized form})$$

4. Tensor Structure and Covariance

Full Covariant T0 Formulation

The complete metric formulation of T0:

$$g_{\mu\nu} = \eta_{\mu\nu} + \frac{2\xi}{E_P} \left(E_{\text{field}} \eta_{\mu\nu} + \frac{\partial_\mu E_{\text{field}} \partial_\nu E_{\text{field}}}{\Lambda^2} \right)$$

where Λ is an energy scale (typically $\Lambda \sim E_P$).

This tensor fulfills:

- ✓ Symmetry: $g_{\mu\nu} = g_{\nu\mu}$
- ✓ Lorentz covariance: Transforms correctly under Lorentz transformations
- ✓ Reduces to Minkowski for $E_{\text{field}} \rightarrow 0$: $g_{\mu\nu} \rightarrow \eta_{\mu\nu}$
- ✓ Generates Riemannian geometry: Non-trivial Christoffel symbols and curvature

Christoffel symbols calculated:

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$$

Riemann tensor calculated:

$$R_{\sigma\mu\nu}^\rho = \partial_\mu \Gamma_{\nu\sigma}^\rho - \partial_\nu \Gamma_{\mu\sigma}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\nu\sigma}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\mu\sigma}^\lambda$$

Explicitly for the T0 metric:

$$R_{\sigma\mu\nu}^\rho = \frac{2\xi}{E_P \Lambda^2} (\partial_\mu \partial_\nu E_{\text{field}} \delta_\sigma^\rho - \partial_\mu \partial_\sigma E_{\text{field}} \delta_\nu^\rho + \text{permutations}) + \mathcal{O}(E_{\text{field}}^2)$$

Non-zero! ✓ Riemannian curvature present.

Ricci tensor:

$$R_{\mu\nu} = R_{\mu\rho\nu}^\rho = \frac{2\xi}{E_P \Lambda^2} (\square E_{\text{field}} \eta_{\mu\nu} - \partial_\mu \partial_\nu E_{\text{field}}) + \mathcal{O}(E_{\text{field}}^2)$$

(Note of 30 Sept. 2026: The part of the metric linear in E_{field} , $(2\xi/E_P) E_{\text{field}} \eta_{\mu\nu}$, produces curvature of order $(\xi/E_P) \partial\partial E_{\text{field}}$ without $1/\Lambda^2$; the term $\partial_\mu E \partial_\nu E / \Lambda^2$ is quadratic in E_{field} and belongs to $\mathcal{O}(E_{\text{field}}^2)$. The prefactor $2\xi/(E_P \Lambda^2)$ in the Riemann and Ricci tensors is therefore not correct.)

Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi GT_{\mu\nu}$$

with the T0 energy-momentum tensor:

$$T_{\mu\nu} = \xi(\partial_\mu E_{\text{field}}\partial_\nu E_{\text{field}} - \frac{1}{2}\eta_{\mu\nu}(\partial E_{\text{field}})^2) + \frac{\lambda}{4}E_{\text{field}}^4\eta_{\mu\nu}$$

5. Magnetic Monopoles – Topological Clarification

Objection

"Stokes' theorem does not apply at singularities"

CORRECT: T0 Allows Topological Monopoles

The T0 statement was **simplified**. Complete version:

Without topological defects:

$$\oint_{\partial V} (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A} = \int_V \nabla \cdot (\nabla \times \vec{E}_{\text{field}}) dV = 0$$

since $\nabla \cdot (\nabla \times \vec{v}) = 0$ for any vector field $\vec{v}$.

With topological defects (monopoles):

For a sphere S^2 around the origin:

$$\oint_{S^2} (\nabla \times \vec{E}_{\text{field}}) \cdot d\vec{A} = 2\pi n \cdot \xi \cdot E_{\text{char}}$$

where $n \in \mathbb{Z}$ is the **topological charge** (winding number) and E_{char} is a characteristic energy scale.

This reproduces Dirac quantization:

The electromagnetic field strength in T0:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + \xi \epsilon_{\mu\nu\rho\sigma} E_{\text{field}} \partial^\rho E_{\text{field}}$$

Magnetic charge:

$$q_m = \oint_{S^2} \vec{B} \cdot d\vec{A}$$

(Corrected on 30 Sept. 2026: previously $q_m = \frac{1}{4\pi} \oint \vec{B} \cdot d\vec{A}$ – with q_m as flux/ 4π the Dirac condition reads $q_e q_m = n\hbar/2$ (Gaussian: $eg = n\hbar c/2$); the form $q_e q_m = 2\pi n\hbar$ used below holds for the full flux $q_m = \oint \vec{B} \cdot d\vec{A}$.)

Dirac quantization condition:

$$q_m q_e = 2\pi n\hbar$$

with the T0 identification:

- Electric charge: $q_e = \xi \cdot E_{\text{char}}$
- Magnetic charge: $q_m = \frac{2\pi n}{\xi}$

Substituting:

$$q_m q_e = \frac{2\pi n}{\xi} \cdot \xi E_{\text{char}} = 2\pi n E_{\text{char}}$$

For $E_{\text{char}} = \hbar$ (in natural units):

$$\boxed{q_m q_e = 2\pi n\hbar} \quad \checkmark$$

(Note of 30 Sept. 2026: With $E_{char} = \hbar = 1$ one would get $q_e = \xi = 1.33 \times 10^{-4}$, whereas $e = \sqrt{4\pi\alpha} = 0.303$ (Heaviside-Lorentz; with the convention $\alpha = 1$: $\sqrt{4\pi} = 3.54$). Moreover $\hbar$ is not an energy; in the corpus $E_{char} = 28.8 \text{ MeV}$.)

Topological interpretation:

The monopole solution corresponds to a map:

$$\phi : S^2 \rightarrow U(1) \cong S^1$$

classified via the transition function on the equator $S^1 \rightarrow U(1)$ with $\pi_1(U(1)) = \mathbb{Z}$ (equivalently $\pi_2(S^2) = \mathbb{Z}$). (Corrected on 30 Sept. 2026: previously "with homotopy group $\pi_2(S^1) = \mathbb{Z}$ " – $\pi_2(S^1) = 0$, since the universal cover $\mathbb{R}$ is contractible; all maps $S^2 \rightarrow S^1$ are therefore trivial.) The winding number n classifies topologically distinct solutions.

Result: T0 contains magnetic monopoles as topological excitations but explains why they are **experimentally rare** (high energy threshold $\sim E_P/\xi$).

6. Quantum Mechanics Integrated

T0 IS a Quantum Field Theory

Canonical quantization of the T0 field:

Field operator:

$$\hat{E}_{\text{field}}(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} (\hat{a}_k e^{ikx} + \hat{a}_k^\dagger e^{-ikx})$$

with:

$$\omega_k = \sqrt{\vec{k}^2 + m_{\text{eff}}^2}, \quad m_{\text{eff}} = \xi \langle E_{\text{field}} \rangle^2$$

Commutation relations:

$$[\hat{a}_k, \hat{a}_{k'}^\dagger] = (2\pi)^3 \delta^3(\vec{k} - \vec{k}')$$

$$[\hat{a}_k, \hat{a}_{k'}] = [\hat{a}_k^\dagger, \hat{a}_{k'}^\dagger] = 0$$

In position space:

$$[\hat{E}_{\text{field}}(t, \vec{x}), \hat{\Pi}(t, \vec{y})] = i\delta^3(\vec{x} - \vec{y})$$

with the conjugate momentum:

$$\hat{\Pi}(x) = \frac{\partial \mathcal{L}}{\partial(\partial_0 \hat{E}_{\text{field}})} = 2\xi \partial_0 \hat{E}_{\text{field}}(x)$$

These are standard quantum field commutation relations!

Particles = Excitations:

- Vacuum state: $|0\rangle$ with $\hat{a}_k|0\rangle = 0$ for all k
- One-particle state: $|k\rangle = \hat{a}_k^\dagger|0\rangle$
- n -particle state: $|n_k\rangle = \frac{(\hat{a}_k^\dagger)^n}{\sqrt{n!}}|0\rangle$ (Fock states)

Specific particle identification:

- Electron: $n = 1, k = k_e, m_e = 0.511 \text{ MeV}$ from the lepton ladder $m_e = \frac{4}{3}\xi^{3/2}v$ (Doc. 006) (Corrected on 30 Sept. 2026: previously " $m_e = \xi E_0^2$ with $E_0 = 0.511 \text{ MeV}$ " – $\xi \cdot 0.511^2 = 3.48 \times 10^{-5} \text{ MeV}^2$, not 0.511 MeV (factor $\approx 1/14700$, wrong dimension); ξE_0^2 with $E_0 = 7.398 \text{ MeV}$ is α .)
- Photon: $n = 1, k = k_\gamma, m_\gamma = 0$ (Goldstone boson of broken symmetry)
- Higgs boson: Excitation around the vacuum expectation value $\langle E_{\text{field}} \rangle = v$

S-matrix and scattering amplitudes:

The scattering matrix is calculated via:

$$S = T \exp \left(-i \int d^4x \mathcal{H}_{\text{int}}(x) \right)$$

with interaction Hamiltonian:

$$\mathcal{H}_{\text{int}} = \frac{\lambda}{4} \hat{E}_{\text{field}}^4$$

Feynman rules:

- Propagator: $\frac{i}{k^2 - m_{\text{eff}}^2 + i\epsilon}$
- Vertex: $-i\lambda$ for E^4 coupling
- ξ -dependent corrections for derivative couplings

7. Empirical Predictions (derived from ξ)**Neutrino masses:**

$$m_\nu = \frac{\xi^2}{2} \cdot m_e \approx 4.54 \text{ meV} \quad \Rightarrow \quad \Delta m_{21}^2 \sim 10^{-5} \text{ eV}^2$$

Note: The formula $m_\nu = \xi^2/2 \cdot m_e$ yields a neutrino mass consistent with cosmological upper bounds. The mass differences from neutrino oscillation data ($\Delta m_{21}^2 \approx 7.5 \times 10^{-5} \text{ eV}^2$) are reproduced in order of magnitude only, (Note of 30 Sept. 2026: $m_\nu^2 = (4.54 \text{ meV})^2 = 2.06 \times 10^{-5} \text{ eV}^2$ lies a factor 3.6 below Δm_{21}^2 ; $\Delta m_{31}^2 \approx 2.5 \times 10^{-3} \text{ eV}^2$ requires a mass $\geq 50 \text{ meV}$, incompatible with the single scale 4.54 meV.) but the mass eigenstates have not yet been fully derived within the T0 framework.

Cosmological constant:

$$\Lambda_{\text{cosmo}} = \frac{\lambda}{4} \langle E_{\text{field}} \rangle^4 \sim (10^{-3} \text{ eV})^4$$

(Note of 1 Oct. 2026: With $\lambda = 2\xi^2$ (from $\lambda/(2\xi) = \xi$, field-equation section) and $\langle E_{\text{field}} \rangle = v = 246 \text{ GeV}$ (Higgs entry above) one obtains $(\lambda/4)^{1/4} v = (\xi^2/2)^{1/4} v = 9.71 \times 10^{-3} \cdot 246 \text{ GeV} \approx 2.4 \text{ GeV}$, hence $\Lambda_{\text{cosmo}} \approx (2.4 \text{ GeV})^4$ and not $(10^{-3} \text{ eV})^4$ – a discrepancy of about 12 orders of magnitude in the energy scale. The formula does not yield the observed value; the table entry below quotes the observed value, not the result of this formula.)

Observable	T0 Prediction	Experimental	Status
Neutrino masses Δm_{21}^2	$m_\nu^2 = 2.1 \times 10^{-5} \text{ eV}^2$	$7.5 \times 10^{-5} \text{ eV}^2$	Order of magnitude (factor 3.6)
Cosmological constant	$\approx (2.4 \text{ GeV})^4$ (formula above)	$(2.1 \times 10^{-3} \text{ eV})^4$	not reproduced
Hubble constant H_0	66.2 km/s/Mpc	see Document 026	2.4σ (Planck), 6.8σ (SH0ES)
Dark matter density Ω_{DM}	0.265	0.264 ± 0.006	✓Consistent

Table 1: Empirical predictions of T0 theory (all without free parameters!)

(Corrected on 1 Oct. 2026: previously Δm_{21}^2 prediction $7.5 \times 10^{-5} \text{ eV}^2$ ✓Consistent, cosmological constant $(2.1 \times 10^{-3} \text{ eV})^4$ ✓Exact, H_0 ✓0.7 σ – the formula $m_\nu = (\xi^2/2) m_e = 4.54 \text{ meV}$

gives $m_\nu^2 = 2.06 \times 10^{-5} \text{ eV}^2$, a factor 3.6 below Δm_{21}^2 (see note above); the Λ row quoted the measured value, while the formula above gives $\approx (2.4 \text{ GeV})^4$; 66.2 km/s/Mpc lies 2.4σ below Planck (67.4 ± 0.5) and 6.8σ below SH0ES (73.0 ± 1.0), see Doc. 026.)

8. Mathematical Consistency Checks

Energy-momentum conservation:

$$\partial_\mu T^{\mu\nu} = 0 \quad \text{satisfied for T0 Lagrangian density}$$

Causality: Light-cone structure from $g_{\mu\nu} \rightarrow$ no superluminal signals.

Unitarity: $S^\dagger S = 1$ for S-matrix, ensured by positive norm in Fock space.

Renormalizability: Dimension of E^4 term: $[E^4] = E^4$, in 4D: $[d^4x] = E^{-4} \rightarrow$ dimensionless coupling parameter $\lambda \rightarrow$ renormalizable.